

SENTRY-Node - Breadboard Lab Manual v9 (Protobloc 2C - rails as-built)

v9 - RAILS AS-BUILT	Per block, SIX rail nets: a continuous BLACK GND line outermost top and bottom, plus FOUR independently named half-rails - V1 (top-left), V2 (top-right), V3 (bottom-left), V4 (bottom-right) - split at mid-block. V1 and V2 are different nets, not one broken rail. Only TWO bridges exist in the whole build (B2 and B3, V1-V2). B3 V3 V4 deliberately carry different signals (SCK WS). Everything else as v8: three vertical channels per block, blocks electrically separate, aux strips at the bottom, posts top-left, DevKit USB-up in B2, e-paper off-board, array coordinates unchanged.
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1 · Board anatomy (verified)



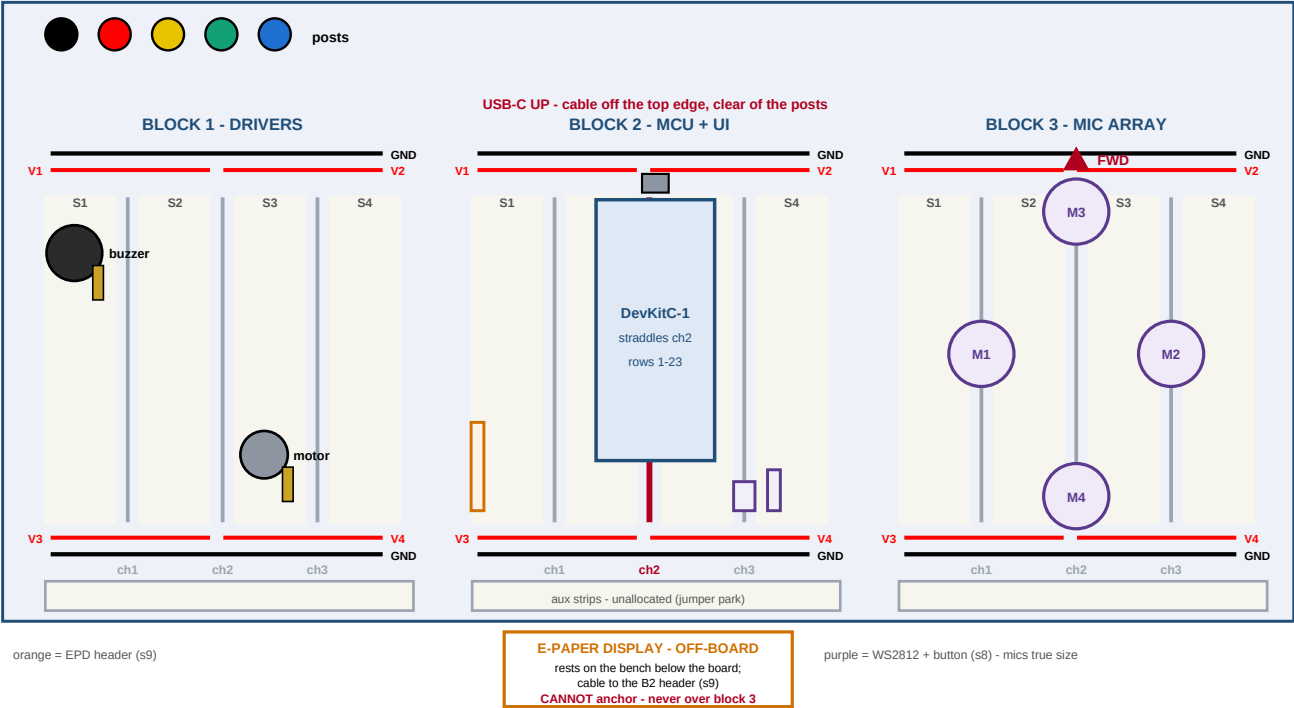
Rails are HORIZONTAL and PER BLOCK - nothing continues from one block to the next.
BLACK rails (outermost) run the whole block = GND. The inner red line is TWO separate named half-rails: V1|V2 (top), V3|V4 (bottom).

Blocks	Three electrically separate blocks side by side. Per block: four vertical sets S1-S4, each 6 hole-columns x 28 rows. Hole columns c1-c24: S1=c1-6, S2=c7-12, S3=c13-18, S4=c19-24.
Nets	One net = one ROW within one set = 6 holes in a horizontal line. Every row has FOUR separate nets (one per set). A seated pin leaves 5 free holes on its net, same row, same side.
Channels	ch1 (c6 c7), ch2 (c12 c13), ch3 (c18 c19) - vertical, 0.3 in across (F2: verify on one module). Two-row parts straddle them; nothing conducts across a channel.
Rails	SIX nets per block, all horizontal. Outermost top + bottom: BLACK continuous GND lines, 24 holes, full block width. Inner top: V1 (left 12 holes) and V2 (right 12) - separate nets. Inner bottom: V3 (left) and V4 (right) - separate nets. The split sits at mid-block, under ch2. NOTHING joins block to block except your jumpers.
Lettering	Board silkscreen: sets read A-F / G-L / A-F / G-L = S1-S4 (c1-24); rows numbered 1-28 on both edges; aux strip a-f x 1-30. V1-V4 are printed at the rail halves.
Aux strips	The "long row" along the board bottom: ~30 columns x 6 per block, no channel. Unallocated - use as a jumper park / spare field.
Posts	5 x 4 mm binding posts, top-left. BLACK is wired to GND as the DMM / scope reference. Others unused.

2 · Census (10 minutes, before anything else)

2a	Count: 3 channels per block, 28 rows per set, 6 holes per row-net. Confirm the aux strip length (expect ~30).
2b	DMM the rails: BLACK top + bottom lines continuous end to end; V1-to-V2 OPEN (separate nets); V3-to-V4 OPEN. Note any block that differs.
2c	Hole-to-hole across each channel: must be 0.3 in / 7.62 mm (one INMP441 module must drop straight in). Check the MIDDLE channel (ch2, the LJA boundary) especially - M3, M4 and the DevKit all straddle it. If 0.2 in anywhere, STOP - flag it.
2d	Seat the DevKit once in block 2 straddling ch2, USB toward the top edge. Record: which columns its two pin rows land in (expect c9/c10 and c17/c18), the row of pin 1, the total row span (expect ~23), and the ROW NUMBER of every labelled pin used in §5.
2e	If any count disagrees with this manual: stop, send the counts, geometry gets reissued from reality.

3 · Whole-board allocation



3.1 · Inter-block wiring (the only things that cross blocks)

Run	From	To	Colour
5V	DevKit 5V pin	B1.V4 (bottom-right half - motor side). No bridge	red
3V3	DevKit 3V3 pin	B2.V1 -> bridge -> B2.V2; B2.V1 -> B1.V1; B2.V2 -> B3.V1 -> bridge -> B3.V2	orange
GND	DevKit GND pin	B2 blacks (top + bottom) -> B1 blacks -> B3 blacks -> BLACK post	black
Clocks	GPIO4 / GPIO5	B3.V3 = SCK (bottom-LEFT half) / B3.V4 = WS (bottom-RIGHT half). NO bridge between them	yellow / white, twisted with a spare GND
SD bus A	GPIO6	M1 pad row -> daisy -> M2 pad row	blue
SD bus B	GPIO7	M3 pad row -> daisy -> M4 pad row	green
Drivers	GPIO17 / GPIO18	B1 buzzer 1k / B1 motor 1k	grey

3.2 · Rail map - all eighteen rail nets

Block	GND-T (black, full)	V1 (top-L)	V2 (top-R)	V3 (bot-L)	V4 (bot-R)	GND-B (black, full)
B1 drivers	GND	3V3 - buzzer	spare	spare	5V - motor. ONLY 5V on the board	GND
B2 MCU+UI	GND (feed + post)	3V3 (feed)	3V3 - BRIDGE to V1	spare	spare	GND - all UI grounds
B3 array	GND	3V3	3V3 - BRIDGE to V1	SCK + tape	WS + tape. NEVER bridge V3-V4	GND

4 · Mic array geometry (values unchanged from v7)

Mic	Pads (block 3)	Centre	x (m)	y (m)	SD	L/R
M1 West	c6 / c7, rows 13-15	(ch1, r14)	-0.02032	0.00000	GPIO6	GND
M2 East	c18 / c19, rows 13-15	(ch3, r14)	+0.02032	0.00000	GPIO6	3V3
M3 North = FWD	c12 / c13, rows 1-3	(ch2, r2)	0.00000	+0.03048	GPIO7	GND
M4 South	c12 / c13, rows 25-27	(ch2, r26)	0.00000	-0.03048	GPIO7	3V3

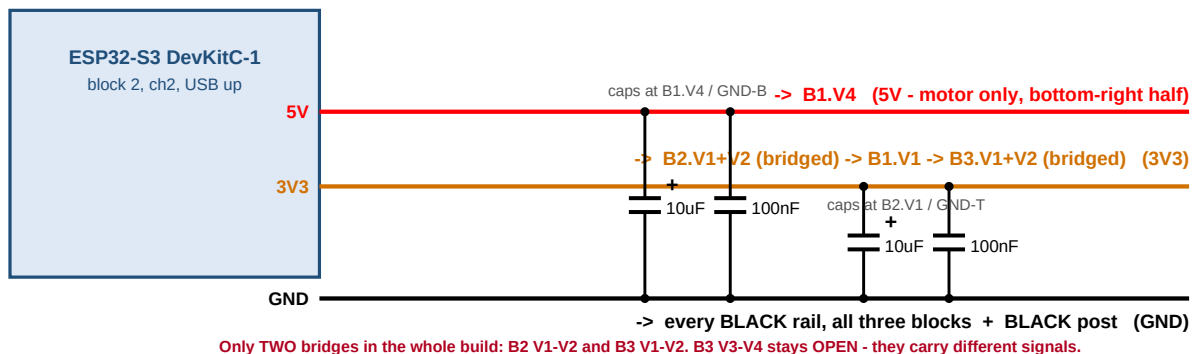
Baselines	E-W = ch1 to ch3 = 16 pitches = 40.64 mm EXACT. N-S = 24 rows on ch2 = 60.96 mm EXACT. Centres coincide at (ch2, row 14). Zero measured constants.
Aliasing	E-W 4220 Hz, N-S 2813 Hz, adjacent arms 4682 Hz. The full 150 Hz - 2 kHz search band is alias-free; the old 1.5 kHz spatial band limit stays DELETED. Max inter-mic delay 178 us.
Orientation	FORWARD = 0 deg = the row-1 edge = the USB edge. The device points the way the cable leaves.
Buses	M1+M2 = bus A (GPIO6), M3+M4 = bus B (GPIO7): each orthogonal pair lives inside one I2S peripheral, so residual clock offset between peripherals cannot corrupt a baseline.

5 · Master pin map

Signal	GPIO	Colour	Notes
I2S SCK - all 4 mics	4	yellow	shared, to B3.V3
I2S WS - all 4 mics	5	white	shared, to B3.V4
I2S SD bus A - M1, M2	6	blue	M1 left slot, M2 right slot
I2S SD bus B - M3, M4	7	green	2nd I2S peripheral, same clock master
E-paper CS/DIN/CLK/DC/RST/BUSY	10/11/12/13/14/15	purple	wire by HAT labels only
WS2812 data	16	grey	via 330R
Buzzer base	17	grey	via 1k
Motor base	18	grey	via 1k
Snooze button	21	grey	to GND, internal pull-up
KEEP CLEAR	0, 3, 19, 20, 26-37, 43-46	-	strapping / USB / flash / PSRAM / UART
Spares	1, 2, 8, 9, 39-42, 47	-	38 / 48 may drive the onboard RGB

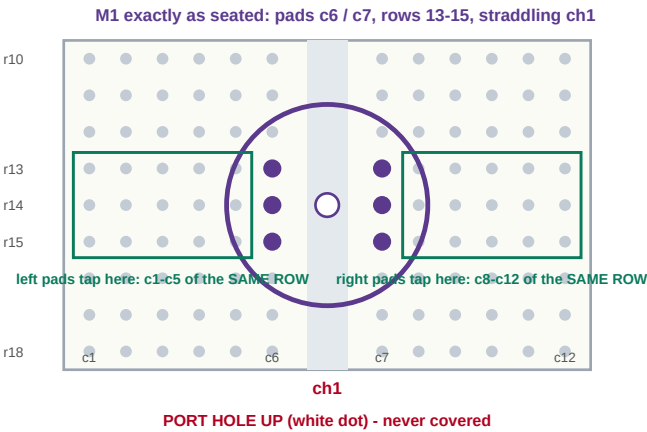
6 · Power [B2 rails + B1 rails]

USB-C from Mac = only power source



Parts	Checks (gate - nothing else is wired until these pass)
10uF x2, 100nF x2, 470uF x1, 2 bridges (B2 + B3 V1-V2), inter-block jumpers per §3.1	3V3 3.25-3.40 V at B1.V1, B2.V1+V2, B3.V1+V2 • 5V 4.6-5.15 V at B1.V4 • every black rail to the black post < 1 ohm • B3.V3 to B3.V4 reads OPEN • electrolytic stripe = GND side

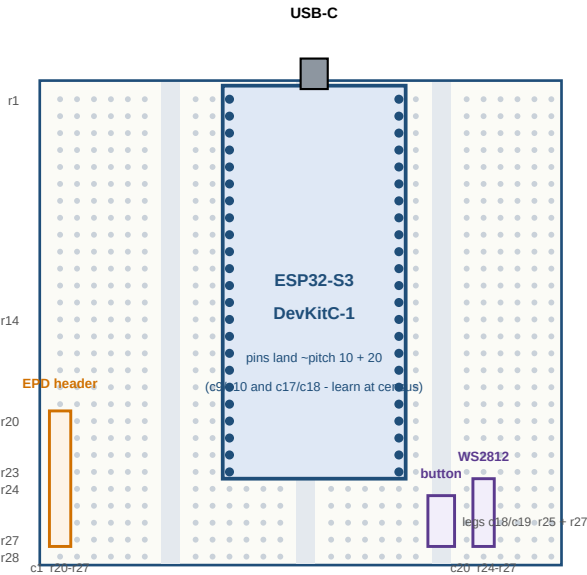
7 · Mic M1 - first light (= Stage 1b) [B3, ch1, pads c6/c7, rows 13-15]



Each ROW-in-set is one 6-hole net. A pad in c6 r13 owns c1-c5 of r13; a pad in c7 r13 owns c8-c12 of r13. Same pattern at ch2 / ch3 (shift columns by +6 / +12).

Seat	M1 straddles B3 ch1, pads c6/c7 in rows 13-15, port up. FINAL position - no relocation later.
Wiring	VDD (c6 r14) -> B3.V1 (3V3) • GND (c6 r15) -> GND-T (black) • SD (c6 r13) -> GPIO6 • SCK (c7 r13) -> B3.V3 (bottom-LEFT half - short run) • WS (c7 r14) -> B3.V4 (bottom-RIGHT half - crosses mid-block) • L/R (c7 r15) -> GND-B (black). Pad-to-signal order is by YOUR module silkscreen - confirm before wiring, variants differ.
Checks	Live 24-bit stream • clap = clean full-scale transient • no stuck or repeated words • steady noise floor

8 · DevKit + UI [B2]



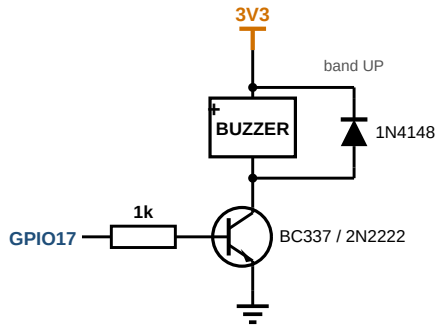
BLOCK 2. DevKit rows 1-23 (23-row span typical - RECORD the real span and the row of every labelled pin at census). Rows 24-28 stay free for the UI.

DevKit	Straddles ch2, rows 1-23, USB-C at the row-1 end, cable off the TOP edge - same side as the posts but two blocks away from them. Never seat it USB-down.
Button	Straddles ch3, legs c18/c19 at rows 25 and 27. GPIO21 -> c13 r25 (S3 net) • c24 r27 -> GND-B (bottom black). Uses one leg from each DIAGONAL pair. 100nF across c13 r25 / c24 r27 if it double-fires.
WS2812	Pins vertical in c20, rows 24-27 (one row-net per pin, identify DIN/VCC/GND by silkscreen). GPIO16 -> c24 r22 -> 330R (c21, r22-r24) -> DIN row. VCC row -> B2.V2 (3V3, right half - nearby), GND row -> GND-B (bottom black). 100uF legs into the VCC and GND rows at c19, + to VCC.

9 · E-paper display [header on B2 - display OFF-BOARD]

HAT pin	VCC	GND	DIN	CLK	CS	DC	RST	BUSY
Header row (c1)	r20	r21	r22	r23	r24	r25	r26	r27
Goes to	B2.V1	GND-B	GPIO11	GPIO12	GPIO10	GPIO13	GPIO14	GPIO15
Anchor rule	The display CANNOT anchor into a breadboard - it has no pins, only a female cable header. Only the 8-pin male strip seats on the board (c1, rows 20-27, one signal per row-net). The display rests on the bench off the BOTTOM edge, screen up, connected by its supplied cable. Never lay it over block 3.							
Rule	Wire by the labels on the HAT and the cable heat-shrink, NEVER by wire colour - WaveShare colour order varies by batch.							
Check	Demo refresh draws • BUSY toggles during refresh • M1 noise floor unchanged while refreshing							

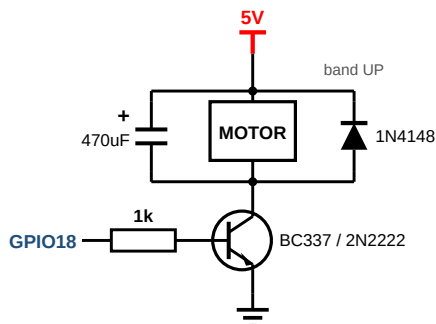
10 · Buzzer driver [B1, S1, rows 4-9]



GPIO HIGH = on. Load between supply and collector, emitter to GND. Check pinout: BC337 and 2N2222 differ.

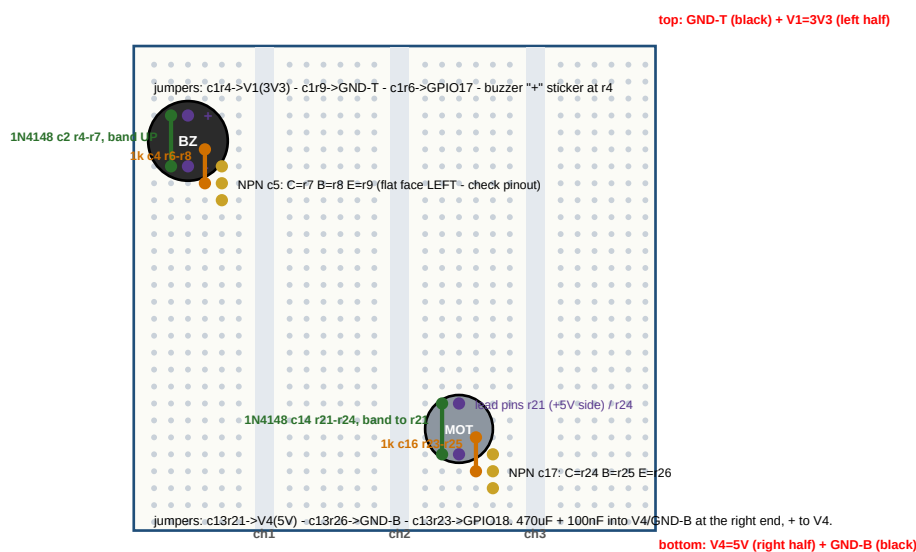
Part ID	Black cylinder, two stiff pins, "+" sticker (3 V active buzzer)
Seat	Buzzer pins c3 r4 (+) / c3 r7 • 1N4148 c2 r4-r7, band UP • NPN c5: C r7, B r8, E r9 • 1k c4 r6-r8
Wire	c1 r4 -> B1.V1 (3V3, top-left half - right there) • c1 r9 -> GND-T (black) • c1 r6 -> GPIO17
Check	Beeps on a test pulse • ~0.7 V base-emitter when on • silent and cool when off

11 · Motor driver [B1, S3, rows 20-27]



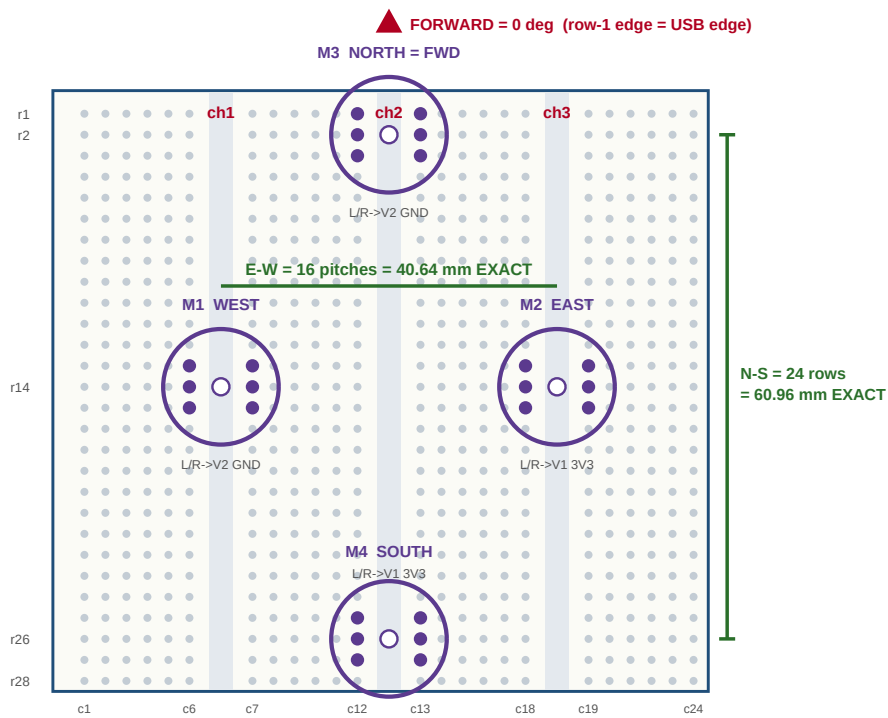
GPIO HIGH = on. Load between supply and collector, emitter to GND. Check pinout: BC337 and 2N2222 differ.

Part ID	Coin / pager disc, red + blue flying leads soldered to male pins first
Seat	Lead pins c15 r21 (5V side) / c15 r24 • 1N4148 c14 r21-r24, band to r21 • NPN c17: C r24, B r25, E r26 • 1k c16 r23-r25
Wire	c13 r21 -> B1.V4 (5V, bottom-right half - right there) • c13 r26 -> GND-B (black) • c13 r23 -> GPIO18 • 470uF + 100nF into V4 / GND-B at the right end, + to V4
Check	Spins on a test pulse • motor OFF: M1 noise floor unchanged from §7



BLOCK 1. Buzzer net top-left near V1/V2; motor net bottom-right near V3/V4 - the two nets never share a row band.

12 · Full array [B3]



Centres coincide at (ch2, row 14). Pads: M1 c6/c7 r13-15 - M2 c18/c19 r13-15 - M3 c12/c13 r1-3 - M4 c12/c13 r25-27.

12a	Set B3 rails: V1+V2 = 3V3 (ONE bridge), blacks = GND, V3 = SCK (tape label), V4 = WS (tape label). Do NOT bridge V3-V4.
12b	Seat M2 (c18/c19, r13-15), M3 (c12/c13, r1-3), M4 (c12/c13, r25-27). All four rotated identically, ports up, nothing resting on B3. M1 is in place from §7.
12c	Six jumpers per mic as the M1 pattern in §7, shifted to each mic. L/R: black GND for M1/M3, 3V3 (V1/V2) for M2/M4. SCK lives on the LEFT half (V3), WS on the RIGHT half (V4): M1 crosses right for WS, M2 crosses left for SCK, M3/M4 sit at the split and reach both.
12d	SD runs: GPIO6 -> M1 SD row -> daisy (free holes of the same row) -> M2 SD row. GPIO7 -> M3 SD row -> M4 SD row. Twist each with a spare GND.
12e	Enter the §4 coordinates in firmware - they are unchanged from v7, so if they are already in, touch nothing.

Tap test (mandatory before any beamforming): 4-channel capture running, tap each mic in turn - spike lands on its expected channel (M1 bus A left, M2 bus A right, M3 bus B left, M4 bus B right). Then one sharp clap: all four within ~1 ms. This also validates the shared clock master across the two peripherals.

13 · Full smoke test

Run	4-channel capture + LED + e-paper refresh + commanded buzzer-and-motor alert burst, all at once
Pass	Noise floor unchanged with LED and e-paper active • buzzer/motor mask audio only during the alert • floor returns to baseline after

14 · Parts

From your order	Qty	From York lab stock	Qty
Protonoc 2C / AD-13 board	1	Jumpers male-male (incl. short bridge links)	60+
ESP32-S3 DevKitC-1 N16R8	1 (+1 spare)	BC337 or 2N2222	3
INMP441 round modules	4	1N4148	4
Waveshare 2.13 e-paper HAT V4 + cable	1	1k / 330R resistors	3 / 2
WS2812 breakout	1 of 5	100nF / 10uF / 100uF / 470uF	8 / 3 / 2 / 1
Active buzzer 3 V	1 of 10	Tact buttons, 3-pin male header strips	2 / 8
Vibration motor 3-5 V	1 of 10	DMM, tape, fine marker	1 each
USB-C DATA cable	1	NOT needed: extra breadboards, clips, backing	0

15 · Risk flags

F1	INMP441 quantity: cart line showed 3; four are needed. Verify pack size or reorder. STILL OPEN.
F2	Mic pad-row spacing must be 0.3 in to straddle a channel. Measure one module at census. Gates the array.
F3	Census disagreement = stop. Counts beat this manual; geometry is reissued from your numbers.
F4	Exactly TWO bridges exist: B2 V1-V2 and B3 V1-V2. NEVER bridge B3 V3-V4 (SCK WS - different nets). B1 needs no bridge at all: V1 and V4 are used as independent half-rails.
F5	WS2812 at 3V3 data: some clones need >3.5 V. Swap modules from the 5-pack before changing anything.
F6	E-paper: display cannot anchor - off-board only, never over B3. Wire by HAT / cable labels, never colour.
F7	Port holes up on every mic. Nothing rests on block 3.
F8	USB current: Mac port or powered hub. Motor pulses peak near 0.4 A.
F9	USB-C always at the row-1 / top edge - clear of the posts, and it fixes FORWARD for the array.

16 · Change-log entries for the Master Brief (§14)

- Rail model corrected from the bench photo: per block, SIX rail nets - continuous BLACK GND lines outermost top and bottom, plus four independently NAMED half-rails V1 (top-left), V2 (top-right), V3 (bottom-left), V4 (bottom-right). V1/V2 are separate nets, not one broken rail.
- Rail assignments rebuilt: only two bridges in the build (B2 and B3, V1-V2 = 3V3). B1 uses V1 (3V3, buzzer) and V4 (5V, motor) as natural half-rails - zero bridges. B3 rides SCK on V3 and WS on V4: the two halves of one physical line carry two different signals, and must never be joined.
- Every part now has a hole-exact seat: (block, column, row) for all pads, resistors, diodes, transistors and jumper landings.
- DevKit seats vertically in block 2 straddling ch2, rows 1-23, USB off the top edge (clear of the posts). FORWARD = the USB edge.
- E-paper display moved OFF-BOARD (it cannot anchor into a breadboard); only its 8-pin header lands, one signal per row-net in B2 c1.
- Mic array relocated to block 3 with IDENTICAL coordinates to v7 (+0.02032 m, +0.03048 m): no firmware change. Spatial band limit remains deleted (worst-case aliasing 2813 Hz).